PRODUCT REQUIREMENTS DOCUMENT (PRD)

CUSTOM METRICS & DATA EXPORT — PRODUCT V2

> **Status:** Approved for design; implementation deferred to a Sonnet-class implementer working from the companion tech spec.
>
> **Version positioning.** This is the **V2 product increment**. It does *not* change V1 scope — the V1 non-goals in `docs/prd.md` §2.2 remain accurate for V1. Instead, V2 pulls two items forward from the V1 PRD's own backlog (`§15.2 V3` → CSV export; `§15.3 Long-Term` → custom dashboards/data) into a delivered V2 scope. See §9, *PRD Impact*.
>
> **Scope note.** This document defines two related capabilities requested together:
>
> - **Capability A — Custom Metrics:** let users import and track their own business KPIs (e.g. PayPal's Total Payment Volume, active accounts) alongside SEC-derived financials.
> - **Capability B — Data Export:** let users download any chart series or table as CSV, for *both* custom metrics and the app's built-in metrics.
>
> They share a single theme — **user-owned data portability, in and out of the app on the user's terms** — and a single reversal of prior V1 non-goals, so they are documented as one feature.

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1. PROBLEM & MOTIVATION

1.1 Capability A — Custom Metrics

Public companies routinely report **management-defined, non-financial key metrics** in the MD&A section of their filings that never appear in the structured XBRL financial statements the app ingests. These are the numbers management itself uses to judge how the business is doing.

Concrete example (PayPal, 2025 Annual Report, Item 7 — *Key Metrics*):

Metric	Meaning	Natural unit
Total Payment Volume (TPV)	Value of payments completed on the platform	Currency

Metric	Meaning	Natural unit
Number of payment transactions	Count of payments completed	Count
Active accounts	Accounts that transacted in the trailing 12 months	Count
Payment transactions per active account	Engagement ratio	Number

A sophisticated investor tracking PayPal wants these trended over time next to revenue and free cash flow — but the app has no way to hold them today, because its metric catalog is a **fixed, closed set** of SEC-derived concepts.

The user need: **plug in your own metrics** — a name, a period, and a value — and have the app store and chart them over time, per company. The app is **not** responsible for how those numbers are gathered or whether they are accurate; it is a **data store and a visualization surface** for numbers the user vouches for.

1.2 Capability B — Data Export

The app renders precise numeric data (revenue, free cash flow yield over time, etc.) but offers **no way to get those numbers out** in a numerical format. Users need to:

1. **Validate** displayed figures against an independent source — which requires the *exact* underlying value, not a rounded `$3.83B` display label.
2. **Share** a series or table in a portable, tool-agnostic format (CSV).
3. **Re-use** the data elsewhere (a spreadsheet, a notebook).

This applies **generically** — to the app's built-in metrics (e.g. free cash flow yield over time) just as much as to custom metrics.

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2. GOALS AND NON-GOALS

2.1 Goals

Capability A — Custom Metrics

1. Let users define custom metrics **per company**, each with a name and a unit.
2. Let users **import** custom metric data (period + value) via CSV.
3. Let users **manually enter and edit** individual data points.
4. **Align** each data point to the company's fiscal periods so custom metrics sit on the same time axis as built-in metrics.

5. **Chart** custom metrics over time, and allow **overlaying** a custom metric against a built-in metric on a shared period axis.
6. Keep custom (user-supplied) data **clearly distinct** from authoritative SEC-derived data, so the app's accuracy guarantee is never diluted.

Capability B — Data Export

7. Let users **download any chart series as CSV** (built-in or custom).
8. Let users **download any financial table as CSV**.
9. Export **exact, full-precision** values (not rounded display labels).
10. Work **fully offline**, consistent with the local-first model.

2.2 Non-Goals

- **User-defined *formulas*.** Custom metrics are *imported data*, not computed derivations. There is no expression engine, no formula editor. (This remains a `docs/prd.md` non-goal.)
- **Accuracy responsibility for custom data.** The app does not validate, reconcile, or attach lineage to user-imported values. It stores what it is given.
- **Automatic scraping of KPIs from filings.** The app does not extract custom metrics from MD&A text. The user supplies the numbers.
- **Cross-company custom metrics / shared definitions.** Each custom metric belongs to exactly one company in V1.
- **Cloud sync / sharing of custom metrics between machines or users.**
- **PDF / image export.** Only CSV (numerical) export is in scope. (PDF/image export remain `docs/prd.md` non-goals.)
- **Import formats other than CSV** (XLSX, JSON, API pull) in V1.

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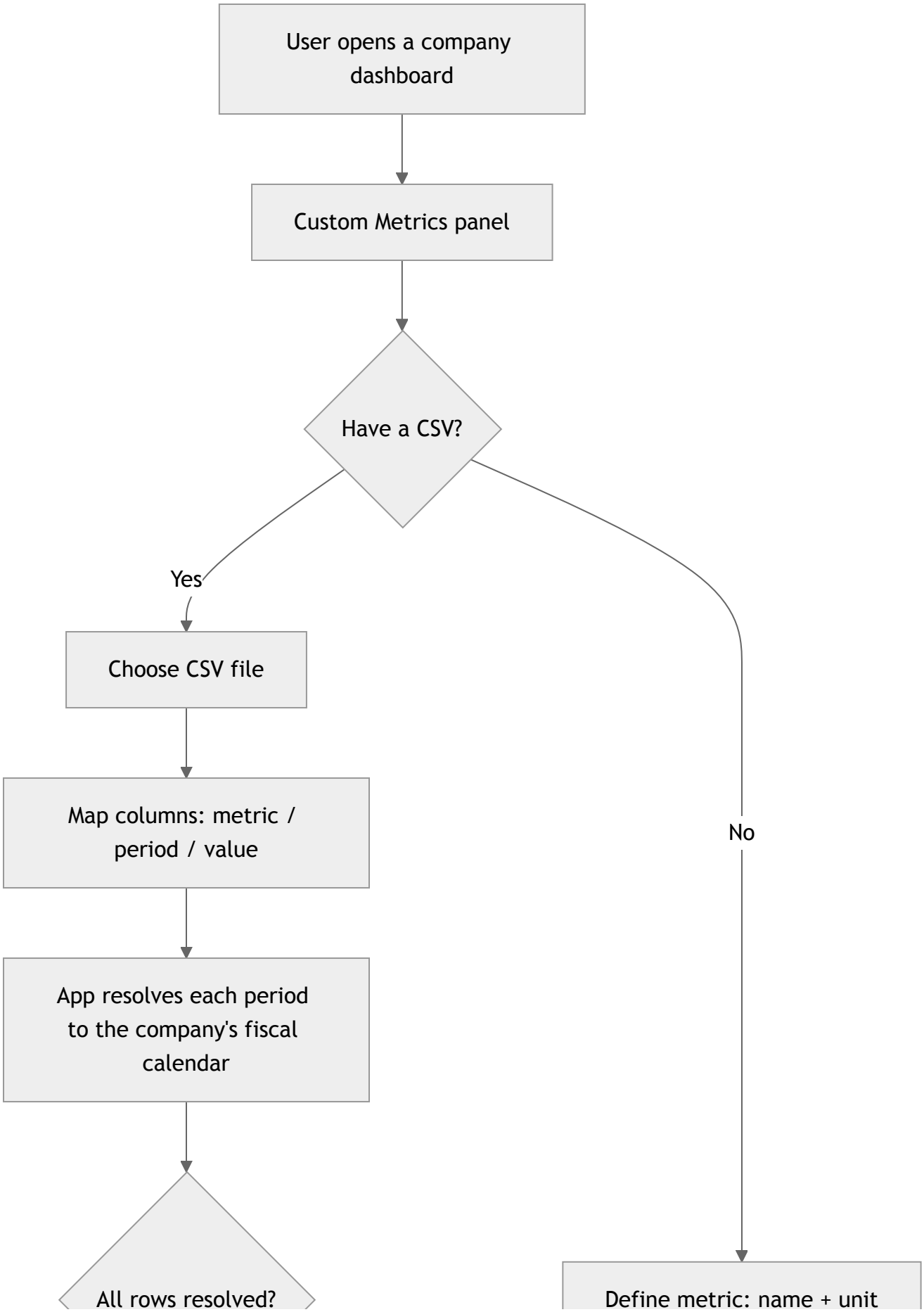
3. USERS & PERSONAS

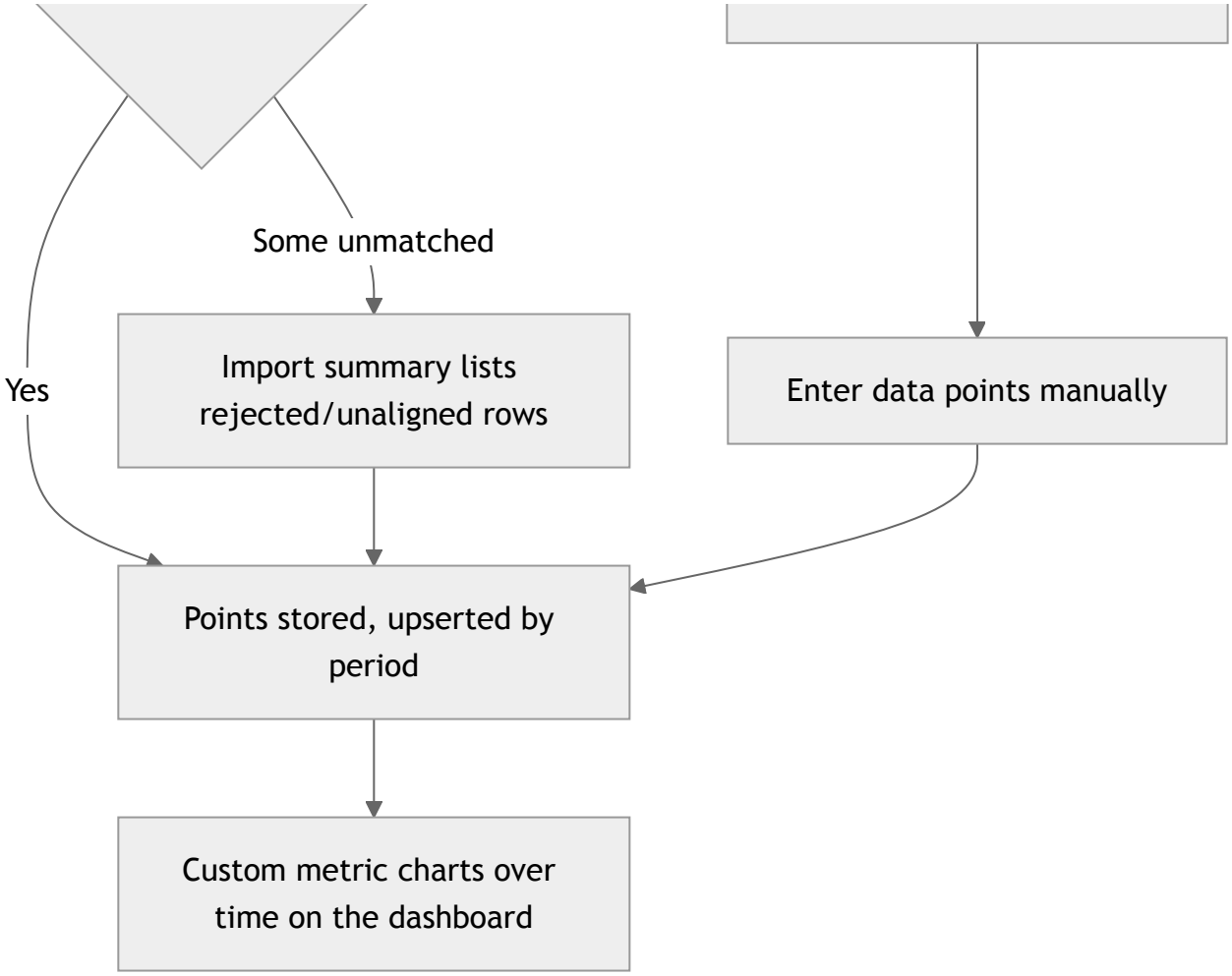
Both capabilities serve the existing personas from `docs/prd.md` §4 — the Long-Term Fundamental Investor and the Independent Financial Researcher — with no new persona. Custom metrics matter most to the fundamental investor who tracks operational KPIs; export matters most to the researcher who wants reproducible, verifiable datasets.

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4. USER WORKFLOWS

4.1 Custom Metric — Define & Import

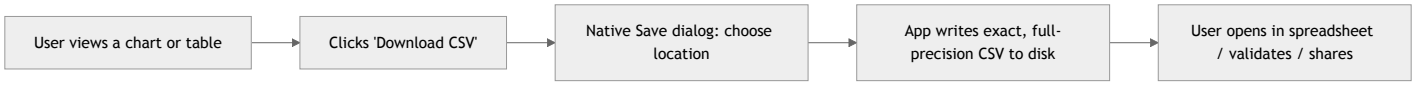

User opens a company dashboard
Custom Metrics panel
Have a CSV?
Yes
No
Choose CSV file
Map columns: metric / period / value
App resolves each period to the company's fiscal calendar
All rows resolved?
Define metric: name + unit



4.2 Custom Metric — Overlay Analysis

1. User selects a built-in metric (e.g. Revenue).
2. User adds one or more custom metrics (e.g. TPV, Active Accounts).
3. App plots all series on a **shared fiscal-period x-axis**, grouping different units onto separate y-axes (currency vs. count vs. percentage).
4. User visually correlates operational KPIs with financial results.

4.3 Data Export — Download a Series or Table



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5. FUNCTIONAL REQUIREMENTS

5.1 Custom Metrics

FR-CM-001 — Define Custom Metric

Users must be able to create a custom metric scoped to a single company, with: a display name, a **unit kind** (currency, count, ratio/percentage, or plain number), and an optional description/definition note.

FR-CM-002 — CSV Import

Users must be able to import custom metric data from a CSV file. The importer must accept a **period identifier** (fiscal label such as `FY2024` or `FY2024 Q2`, **or** a calendar date) and a **value** per row, and may create or update multiple metrics in one file (long format: metric, period, value).

FR-CM-003 — Period Alignment

Each imported/entered data point must be resolved to one of the company's fiscal periods (annual or quarterly). Both annual and quarterly data must be supported for the same metric.

FR-CM-004 — Transparent Import Results

The importer must **never silently discard** rows. Every row is reported as imported, updated, or rejected (with a reason: unparseable, no matching fiscal period, duplicate). Unmatched rows are retained so they can be re-aligned after a later ingestion adds the relevant period. *(Consistent with `docs/prd.md` FR-040.)*

FR-CM-005 — Manual Entry & Editing

Users must be able to add, edit, and delete individual data points and metric definitions in the UI, without a CSV.

FR-CM-006 — Idempotent Re-import

Re-importing data for an existing (metric, period) must **update** the value, not create duplicates.

FR-CM-007 — Charting

Custom metrics must chart over time on the company dashboard, using unit-appropriate formatting (currency, count, percentage, or decimal number).

FR-CM-008 — Overlay With Built-in Metrics

Users must be able to overlay a custom metric against a built-in metric (and against other custom metrics) on a shared fiscal-period x-axis, with distinct y-axes per unit family.

FR-CM-009 — Trust Separation

Custom (user-supplied) metrics must be **visually distinguished** from authoritative SEC-derived metrics and must be **excluded** from the app's source-lineage/traceability guarantees. Custom data must never enter the normalization pipeline or be presented as filing-derived. The most the app records is user-supplied provenance (e.g. source filename, import timestamp).

FR-CM-010 — Deletion Safety

Deleting a company or a custom metric must warn the user when user-authored custom data will be lost, since that data is not re-derivable from filings.

5.2 Data Export

FR-EX-001 — Series Export

Any single-metric time-series chart — built-in or custom — must offer a "Download CSV" action producing one row per period.

FR-EX-002 — Comparison Export

The overlay/comparison chart must export a CSV with one column per series, aligned by period.

FR-EX-003 — Table Export

Financial-statement tables must offer a "Download CSV" action producing the table as shown (metrics × periods).

FR-EX-004 — Full Precision

Exported values must be **exact** — the full-precision underlying number, not the compact display label (e.g. `383285000000.00`, not `$383.29B`). This is essential to the validation use case.

FR-EX-005 — Unit Clarity

Exported CSVs must make units unambiguous: currency values in currency units with a currency column, ratio/percentage metrics expressed as a percentage (and/or decimal), counts as integers.

FR-EX-006 — Period Context

Series exports must include enough period context to be self-describing: the period label, fiscal year, fiscal quarter, and period start/end dates.

FR-EX-007 — Offline

Export must function with no network connection.

FR-EX-008 — What-you-see

Export produces exactly the data shown in the originating view (same metric, same annual/quarterly mode, same periods).

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6. UX REQUIREMENTS

- Consistent with `docs/prd.md` §10: dense, professional, analytical, minimal.
- Custom metrics live in a clearly labelled area of the dashboard, badged as **user-supplied**.
- "Download CSV" is a small, consistent affordance attached to every eligible chart and table — never a modal-heavy flow.
- Import surfaces a clear results summary; errors are explicit and per-row.

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7. SUCCESS CRITERIA

1. A user can import PayPal's four key metrics from a CSV and see each charted over time within one workflow, with any unmatched rows explicitly reported.
2. A user can overlay TPV against Revenue on one chart with correct dual axes.
3. A user can download the free-cash-flow-yield series and the custom TPV series as CSVs whose values match the exact stored numbers.
4. No custom metric is ever presented as SEC-sourced or shown with filing lineage.
5. All of the above work offline.

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8. EDGE CASES

- **Company not yet ingested / no periods:** custom-metric import requires the company's fiscal periods to exist; if none do, the importer reports rows as unaligned and retains them for later re-alignment.
- **Date that spans a period boundary:** resolved deterministically (see design doc); ambiguous rows are rejected with a reason, not guessed.
- **Very large counts / currency values:** exported at full precision from the authoritative integer store (see design doc §4 on the precision guarantee).

- **Re-ingestion changing the fiscal calendar:** stored custom points retain their original observed date and are re-resolved to periods on refresh.
- **Ratio/percentage inputs:** the importer must know whether a value is a decimal ratio (`0.253`) or a percentage (`25.3`); this is a per-metric setting, not guessed per row.

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9. PRD IMPACT (`docs/prd.md`)

This is a **V2 increment**, so the V1 PRD's scope and non-goals stay intact and accurate *for V1*. Per the standing PRD-update protocol, `docs/prd.md` is updated only to record that these previously-backlogged items are now scheduled into V2:

1. **§2.2 Non-Goals:** annotate the *CSV export* and *Custom widgets* lines as "*(V1 non-goal; delivered in V2 — see `docs/custom-metrics-prd.md`)*". The lines themselves remain, because they still describe V1.
2. **§15.1 (V2 Enhancements):** add **Custom metrics (user-imported KPIs)** and **CSV export** as in-scope V2 items, with a pointer to this doc set.
3. **§15.2 (V3):** remove **CSV export** (promoted to V2); **spreadsheet interoperability** and **chart/image export** remain V3.

What stays a non-goal in V2 as well: **user-defined formulas**, a **plugin ecosystem**, **PDF/image export**, and **cross-company / cloud-synced** custom metrics (§2.2 above).

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10. OPEN QUESTIONS

None blocking. Deferred to design:

- Exact CSV column layout for comparison exports (documented in design doc).
- Whether manual entry ships in the same release as CSV import (assumed yes — it is low-cost given the same storage path).